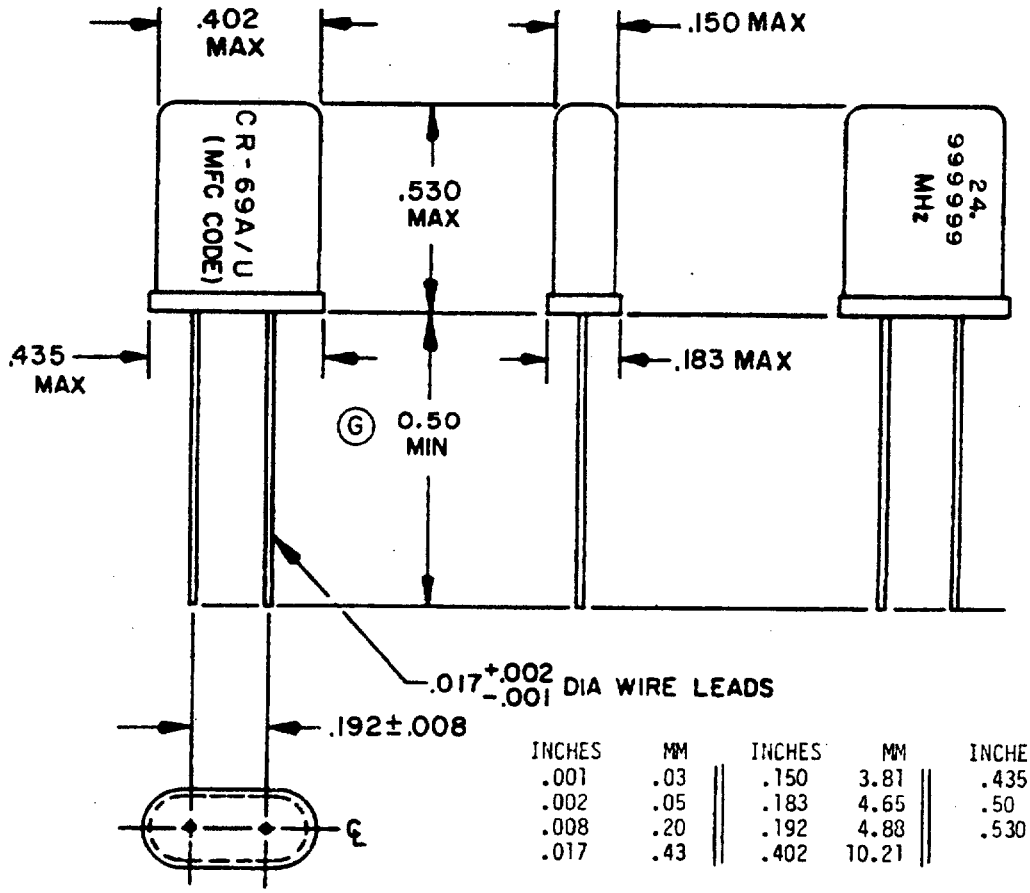
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-69A/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 2.9 to 25 MHz; fundamental; noncontrolled; antiresonance.




| INCHES | MM | INCHES | MM | INCHES | MM |
|---|---|---|---|---|---|
| .001 | .03 | .150 | 3.81 | .435 | 11.05 |
| .002 | .05 | .183 | 4.65 | .50 | 12.7 |
| .008 | .20 | .192 | 4.88 | .530 | 13.46 |
| .017 | .43 | .402 | 10.21 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The type-designation, frequency, and code-designating characters shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.

FIGURE 1. Dimensions and configuration.

Ⓖ denotes changes



FSC 5955

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TABLE I. Requirements.

| Requirements | Specified value or condition |
|---|---|
| Ⓖ Holder | HC-18/U or HC-43/U (MIL-H-10056/9) |
| Dimensions, marking, and configuration | See figure |
| Frequency range | 2.9 to 25 MHz, inclusive |
| Capacitance, shunt | 7 pF, maximum |
| Frequency tolerances: | |
| Primary operating temperature range | ±20 PPM |
| Secondary operating temperature range | +30 PPM |
| Equivalent resistance | See table III |
| Mode of oscillation | Fundamental |
| Antiresonance, load capacitance | 30.0 ±0.5 pF |
| Operating temperature ranges: | |
| Primary | -40°C to +90°C, inclusive |
| Secondary | -55°C to -40°C and +90°C to +105°C, inclusive |
| Test set, calibration values, and rated drive level | See table II |
| Shock: | |
| Frequency change permitted | ±5 PPM |
| Equivalent resistance change permitted | ±10 percent |
| Vibration: 1/ | |
| Frequency change permitted | ±5 PPM |
| Equivalent resistance change permitted | ±10 percent |
| Thermal shock: | |
| Frequency change permitted | ±5 PPM |
| Equivalent resistance change permitted | ±10 percent |
| Aging: | |
| Frequency change permitted | ±5 PPM |

1/ Method 204 of MIL-STD-202, test condition A.

TABLE II. Test set, calibration values, and rated drive level.

| Government reference standard test set | Band of specified frequencies, inclusive | Calibration values | | | Rated drive level |
|---|---|---|---|---|---|
| | | Resistance | Crystal current | Resistor voltage drop | |
| | MHz | Ohms | mA | Volt | mW |
| TS-330/TSM | 2.9 to 5.1 | 50 | 10 | --- | 5.0 ±1.0 |
| | 5.1+ to 7.5 | 22 | 15 | --- | 5.0 ±1.0 |
| | 7.5+ to 10 | 13 | 20 | --- | 5.0 ±1.0 |
| TS-683/TSM | 10+ to 25 | 10 | --- | 0.22 | 5.0 ±1.0 |



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TABLE III. Equivalent resistance.

| Frequency range, inclusive | Maximum resistance |
|---|---|
| MHz | Ohms |
| 2.9 to 3.75 | 180 |
| 3.75+ to 4.75 | 120 |
| 4.75+ to 6 | 75 |
| 6+ to 7 | 50 |
| 7+ to 10 | 30 |
| 10+ to 25 | 25 |

NOTE: NATO preferred type NXT3VA.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

Users activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5955-0623-7)



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